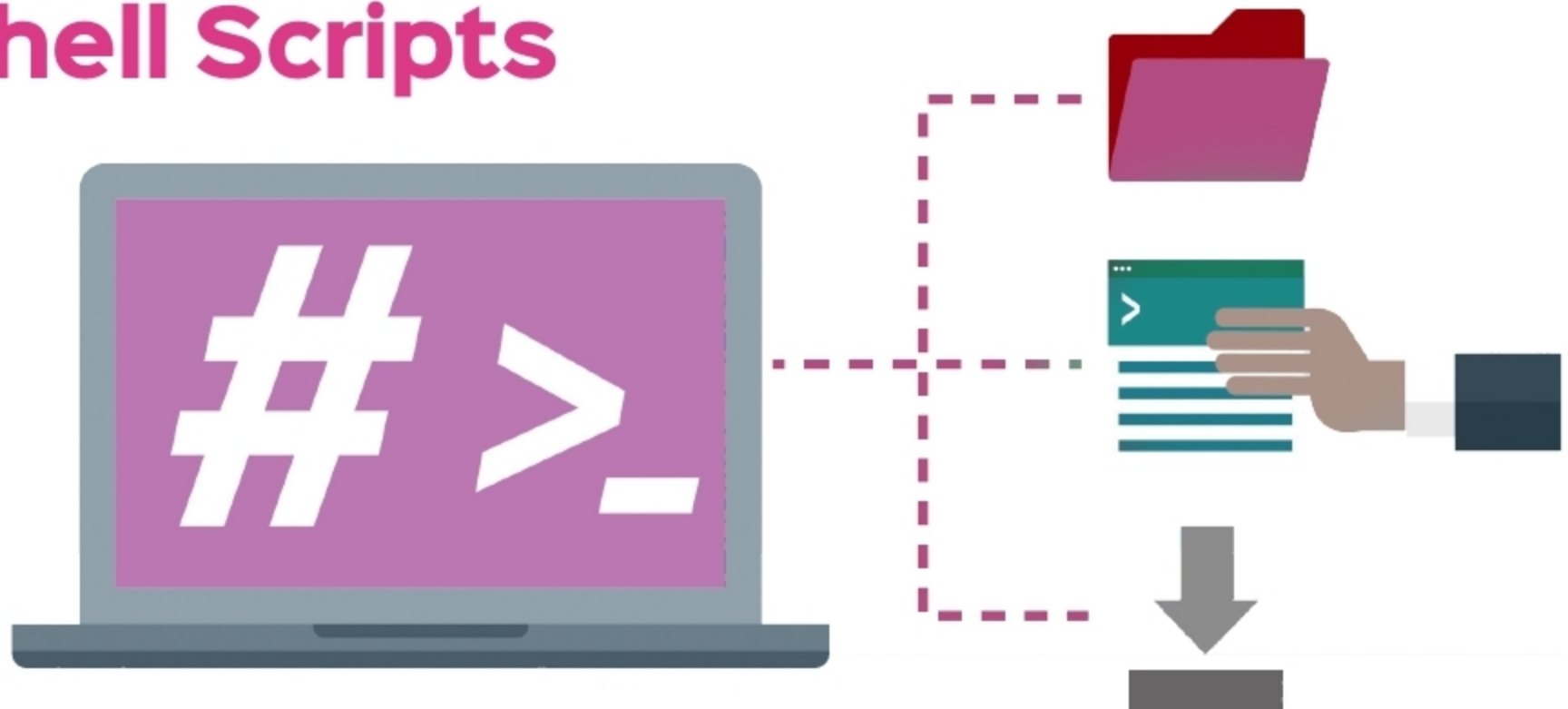


Programming with Shell Scripts



This article will introduce you to the basics of programming with shell scripts.

The best thing about GNU/Linux is that it's packed with utilities that improve productivity immensely. One such utility is *shell* that can help accomplish complex tasks with just a few sequences of commands. Most of the time, users operate shell in an interactive way. However, it becomes really productive when we leverage the scripting capabilities of shell. In this beginner-friendly article we will take a look at shell scripting. We will use bash shell to demonstrate all the examples. Then we will discuss important concepts like variables, conditional expressions, loops, etc, with simple yet meaningful examples.

First, the shell script

One might ask what a shell script is? Well, it is a plain text file that contains a sequence of shell commands. Let us understand this with a simple example. Given below is a simple 'echo' command, which prints a message on standard output:

```
$ echo "Programming with Shell script"
Programming with Shell script
```

Now let us create a simple shell script that displays the above messages on the standard output. We will perform the following steps in sequence to achieve this:

1. Open your favourite text editor.
2. Create a new file, namely 'first_script.sh', and add the following content to it:

```
echo "Programming with Shell script"
```

3. Your file content should look like this:

```
$ cat first_script.sh
```

```
echo "Programming with Shell script"
```

4. Execute shell script as follows:

```
$ bash first_script.sh
```

```
Programming with Shell script
```

Wonderful! You have written your first shell script.

Comments

Comments are used to improve the readability of the script. Shell uses the pound (#) symbol for comments. A line beginning with the pound (#) symbol is ignored by the shell interpreter during execution. The script given below shows the usage of the comment:

```
$ cat first_script.sh
```

```
# Displays message on standard output
echo "Programming with Shell script"
```

This script generates the following output when it is executed:

```
$ bash first_script.sh
Programming with Shell script
```

Note that the output doesn't contain any line beginning with the pound (#) symbol.

